Lect#6: Project Cost Management

What is Cost and Project Cost Management?

- **Cost** is a resource sacrificed or fore-gone to achieve a specific objective or something given up in exchange
- Costs are usually measured in monetary units like dollars
- **Project cost management** includes the processes required to ensure that the project is completed within an approved budget

Project Cost Management Processes

1. **Resource planning: determining what resources and quantities of them should be used**
2. **Cost estimating: developing an estimate of the costs and resources needed to complete a project**
3. **Cost budgeting: allocating the overall cost estimate to individual work items to establish a baseline for measuring performance**
4. **Cost control: controlling changes to the project budget**

Basic Principles of Cost Management

- Most CEOs and boards know a lot more about finance than IT. IT project managers must speak their language
  - Profits are revenues minus expenses
  - Life cycle costing is estimating the cost of a project over its entire life
  - Cash flow analysis is determining the estimated annual costs and benefits for a project
  - Benefits and costs can be tangible or intangible, direct or indirect
  - Sunk cost should not be a criteria in project selection

1. Resource Planning

- The nature of the project and the organization will affect resource planning
- Some questions to consider:
  - How difficult will it be to do specific tasks on the project
  - Is there anything unique in this project's scope statement that will affect resources?
  - What is the organization's history in doing similar tasks?
  - Does the organization have or can they acquire the people, equipment, and materials that are capable and available for performing the work?

2. Cost Estimating

- An important output of project cost management is a cost estimate
- There are several types of cost estimates and tools and techniques to help create them
- It is also important to develop a cost management plan that describes how cost variances will be managed on the project

Table 6-2. Types of Cost Estimates

Type of Estimate	When Done	Why Done	How Accurate
Rough Order of Magnitude (ROM)	Very early in the project life cycle, often 3–5 years before project completion	Provides rough ballpark of cost for selection decisions	–25%, +75%
Budgetary	Early, 1–2 years out	Puts dollars in the budget plans	–10%, +25%
Definitive	Later in the project, < 1 year out	Provides details for purchases, estimate actual costs	–5%, +10%

Cost Estimation Tools and Techniques

- 3 basic tools and techniques for cost estimates:
  - Top-down: use the actual cost of a previous, similar project as the basis for the new estimate
  - Bottom-up: estimate individual work items and sum them to get a total estimate
  - Parametric: use project characteristics in a mathematical model to estimate costs

Typical Problems with IT Cost Estimates

- Developing an estimate for a large software project is a complex task requiring a significant amount of effort. Remember that estimates are done at various stages of the project
- Many people doing estimates have little experience doing them. Try to provide training and mentoring
- People have a bias toward underestimation. Review estimates and ask important questions to make sure estimates are not biased
- Management wants a number for a bid, not a real estimate. Project managers must negotiate with project sponsors to create realistic cost estimates

Table 6-3. Business Systems Replacement Project Cost Estimate Overview

Category	Description
Objective	Install a suite of packaged financial applications software which will enable more timely information for management decision-making, easier access to data by the ultimate end user, and allow for cost savings through productivity improvements throughout the company.
Scope	The core financial systems will be replaced by Oracle financial applications. These systems include: • General Ledger • Fixed Assets • Ops Report [AU: spell out Ops] • Accounts Payable • Accounts Receivable • Project Accounting • Project Management
Assumptions	Oracle's software provides • Minimal customization • No change in procurement systems during accounts payable implementation
Cost/Benefit Analysis & Internal Rate of Return (IRR)	BSR was broken down into a three-year cash outlay without depreciation. Costs are represented in thousands. Capital and expenses are combined in this example.

Table 6-4. Business Systems Replacement Project Cash Flow Analysis

	FY95 ($000)	FY96 ($000)	FY97 ($000)	3 Year Total ($000)	Future Annual Costs/Savings ($000)
Costs					
Oracle/PM Software (List Price)	992	500	0	1492	0
60% Discount	(595)			(595)	
Oracle Credits	(397)	0		(397)	
Net Cash for Software	0	500		500	
Software Maintenance	0	90	250	340	250
Hardware & Maintenance	0	270	270	540	270
Consulting &Training	205	320	0	525	0
Tax & Acquisition	0	150	80	230	50
Total Purchased Costs	205	1330	600	2135	570
Information Services & Technology (IS&T)	500	1850	1200	3550	0
Finance/Other Staff	200	990	580	1770	
Total Costs	905	4170	2380	7455	570
Savings					
Mainframe		(101)	(483)	(584)	(597)
Finance/Asset/PM		(160)	(1160)	(1320)	(2320)
IS&T Support/Data Entry		(88)	(384)	(472)	(800)
Interest		0	(25)	(25)	(103)
Total Savings		(349)	(2052)	(2401)	(3820)
Net Cost (Savings)	905	3821	328	5054	(3250)
8 Year Internal Rate of Return	35%				

3. Cost Budgeting

- Cost budget involves allocating the project cost estimate to individual work items and providing a cost baseline
- For example, in the Business Systems Replacement project, there was a total purchased costs estimate for FY97 of $600,000 and another $1.2 million for Information Services and Technology
- These amounts were allocated to appropriate budgets as shown in Table 6-5

Table 6-5. Business Systems Replacement Project Budget Estimates for FY97 and Explanations

Budget Category	Estimated Costs	Explanation
Headcount (FTE)	13	Included are 9 programmer/analysts, 2 database analysts, 2 infrastructure technicians.
Compensation	$1,008,500	Calculated by employee change notices (ECNs) and assumed a 4% pay increase in June. Overload support was planned at $10,000.
Consultant/Purchased Services	$424,500	Expected consulting needs in support of the Project Accounting and Cascade implementation efforts; maintenance expenses associated with the Hewlett-Packard (HP) computing platforms; maintenance expenses associated with the software purchased in support of the BSR project.
Travel	$25,000	Incidental travel expenses incurred in support of the BSR project, most associated with attendance of user conferences and off-site training.
Depreciation	$91,000	Included is the per head share of workstation depreciation, the Cascade HP platform depreciation, and the depreciation expense associated with capitalized software purchases.
Rents/Leases	$98,000	Expenses associated with the Mach1 computing platforms.
Other Supplies and Expenses	$153,000	Incidental expenses associated with things such as training, reward and recognition, long distance phone charges, miscellaneous office supplies.
Total Costs	**$1,800,000**	

4. Cost Control

- Project cost control includes
  - monitoring cost performance
  - ensuring that only appropriate project changes are included in a revised cost baseline
  - informing project stakeholders of authorized changes to the project that will affect costs
- Earned value analysis is an important tool for cost control

Earned Value Analysis (EVA)

- EVA is a project performance measurement technique that integrates scope, time, and cost data
- Given a baseline (original plan plus approved changes), you can determine how well the project is meeting its goals
- You must enter actual information periodically to use EVA. Figure 6-1 shows a sample form for collecting information

Figure 6-1. Cost Control Input Form for Business Systems Replacement Project

WBS#: 6.8.1.2	Description: Design Interface Process - Customer Information	Revision:	Revision Date:

Assignments

Responsible:	SMC	Role:	PA	Availability:	6
Involved:		Role:		Availability:	
Involved:		Role:		Availability:	
Involved:		Role:		Availability:	

Forecast

	Hours per day	Effort (in hours)
Optimistic:	20	
Most Likely:	30	
Pessimistic:	40	
Delay (Days):		

Calculated

Plan Effort: 30 Hrs

Plan Duration: 5 Days

Description

Develop an operational process design for the Customer Information interface from the Invoicing System to Oracle Receivables. This task will accept as input the business/functional requirements developed during the tactical analysis phase and produce as output a physical operational design, which provides the specifications, required for code development.

Assumptions

- All business rules and issues will be resolved prior to this task.
- The ERD & data model for Oracle Receivables & any Oracle extension required will be completed and available prior to this task.
- The ERD for the Invoicing System will be completed and available prior to this task.
- Few iterations of the review/modify cycle will be required.

Results / Deliverables

Process Design Document - Technical
- Operation/Physical DFD
- Process Specifications
- Interface Data Map

Dependencies

Predecessors (WBS#):				Successors (WBS#):	
4.7					

Earned Value Analysis Terms

- **Budgeted cost of work scheduled** (BCWS), also called the budget, is that portion of the approved total cost estimate planned to be spent on an activity during a given period
- **Actual cost of work performed** (ACWP), also called actual cost, are the total direct and indirect costs incurred in accomplishing work on an activity during a given period
- **Budgeted cost of work performed** (BCWP), also called earned value, is the percentage of work actually completed multiplied by the planned cost (or BCWS)

Table 6-6. Earned Value Calculations for One Activity After Week One

Activity	Week 1	Week 2	Total	% Complete after Week 1	EarnedValue after Week 1 (BCWP)
Purchase web server	10,000	0	10,000	75%	7,500
Weekly Plan (BCWS)	10,000	0	10,000		
Weekly Actual (ACWP)	15,000	5,000	20,000		
Cost Variance (CV)	-7,500				
Schedule Variance (SV)	-2,500				
Cost Performance Index (CPI)	50%				
Schedule Performance Index (SPI)	75%				

Table 6-7. Earned Value Formulas

Term	Formula
Earned Value	Budgeted Cost of Work Performed (BCWP) = budgeted cost to date X % complete
Cost Variance	CV=BCWP-ACWP (actual cost of work performed)
Schedule Variance	SV=BCWP-BCWS (budgeted cost of work scheduled)
Cost Performance Index	CPI=BCWP/ACWP
Schedule Performance Index	SPI = BCWP/BCWS

Rules of Thumb for EVA Numbers

- Negative numbers for cost and schedule variance indicate problems in those areas. The project is costing more than planned or taking longer than planned
- CPI and SPI less than 100% indicate problems

Figure 6-2. Earned Value Calculations for a One-Year Project After Five Months

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	**Activity**	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**	**Plan**	**% Complete**	**EV**
2	Plan and staff project	4,000	4,000											8,000	100	8,000
3	Analyze requirements		6,000	6,000										12,000	100	12,000
4	Develop ERDs			4,000	4,000									8,000	100	8,000
5	Design database tables				6,000	4,000								10,000	100	10,000
6	Design forms, reports, and queries					8,000	4,000							12,000	50	6,000
7	Construct working prototype						10,000							10,000	-	-
8	Test/evaluate prototype						2,000	6,000						8,000	-	-
9	Incorporate user feedback							4,000	6,000	4,000				14,000	-	-
10	Test system									4,000	4,000	2,000		10,000	-	-
11	Document system											3,000	1,000	4,000	-	-
12	Train users												4,000	4,000	-	-
13	Monthly Plan	4,000	10,000	10,000	10,000	12,000	16,000	10,000	6,000	8,000	4,000	5,000	5,000	100,000		**44,000**
14	BCWS or Cumulative Plan	4,000	14,000	24,000	34,000	46,000	62,000	72,000	78,000	86,000	90,000	95,000	100,000			
15	Monthly Actual	4,000	11,000	11,000	12,000	15,000										
16	ACWP or Cumulative Actual	4,000	15,000	26,000	38,000	53,000										
17	Monthly EV	4,000	10,000	10,000	10,000	10,000										
18	BCWP or Cumulative EV	4,000	14,000	24,000	34,000	44,000										
19	Project BCWP as of May 31	44,000														
20	Project BCWS as of May 31	46,000														
21	Project ACWP as of May 31	$ 53,000														
22	**CV=BCWP-ACWP**	**$ (9,000)**														
23	**SV=BCWP-BCWS**	**$ (2,000)**														
24	**CPI=BCWP/ACWP**	**83%**														
25	**SVI=BCWP/BCWS**	**96%**														
26	**Estimated cost at completion**	**$120,455**	(original plan of $100,000 divided by CPI of 83%)													
27	**Estimated time to complete**	**12.55**	(original plan of 12 months divided by SVI of 96%)													

Excel file

Using Software to Assist in Cost Management

- Spreadsheets are a common tool for resource planning, cost estimating, cost budgeting, and cost control
- Many companies use more sophisticated and centralized financial applications software for cost information
- Project management software has many cost-related features